

Some Applications of Trigonometry — MCQ Worksheet

Mathematics • Chapter 9 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The angle of elevation is the angle made by the line of sight with the horizontal when the object is:

- (A) Below horizontal
- (B) Above horizontal
- (C) On horizontal
- (D) Behind viewer

Q2. The angle of depression is the angle made by the line of sight with the horizontal when the object is:

- (A) Above horizontal
- (B) Below horizontal
- (C) On horizontal
- (D) Vertical

Q3. A tower of height $50\sqrt{3}$ m casts a shadow of length 50 m on horizontal ground. The angle of elevation of the sun is:

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Q4. A 1.5 m tall boy stands at distance $\sqrt{3}$ m from a lamp post and casts a shadow of 1 m on the ground. Length of lamp post is:

- (A) 3 m
- (B) 6 m
- (C) 4.5 m
- (D) 2.5 m

Q5. The angle of elevation of the top of a tower from a point 30 m away from the foot is 30° . The height of the tower is:

- (A) 10 m
- (B) $10\sqrt{3}$ m
- (C) 30 m
- (D) $30\sqrt{3}$ m

Q6. From a point 50 m away, the angle of elevation of the top of a building is 45° . The height of the building is:

- (A) 25 m
- (B) 50 m
- (C) 100 m
- (D) 75 m

Q7. From a point on the ground, the angle of elevation of the top of a tower is 60° , and 30 m away from the foot. Height is:

- (A) 30 m
- (B) $30\sqrt{3}$ m
- (C) $10\sqrt{3}$ m
- (D) 60 m

Q8. The angle of depression from the top of a 100 m cliff to a boat is 30° . Distance of boat from the foot is:

- (A) 100 m
- (B) $100\sqrt{3}$ m
- (C) $100/\sqrt{3}$ m
- (D) 50 m

Q9. A ladder 10 m long reaches a window 8 m above the ground. The distance of foot of ladder from wall is:

- (A) 6 m
- (B) 4 m
- (C) 8 m
- (D) 10 m

Q10. A kite is flying at height 60 m above the ground. The string is taut and makes an angle of 60° with the ground. Length of string is:

- (A) 60 m
- (B) $40\sqrt{3}$ m
- (C) $30\sqrt{3}$ m
- (D) $60 \cdot 2/\sqrt{3} = 40\sqrt{3}$ m (≈ 69.3 m)

Q11. If the shadow of a tower is $\sqrt{3}$ times its height, the angle of elevation of the sun is:

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Q12. A man on the deck of a ship 8 m above water level observes a hill at angle of elevation 60° and base at angle of depression 30° . The distance of hill from ship is:

- (A) $8\sqrt{3}$ m
- (B) 16 m
- (C) 12 m
- (D) 8 m

Q13. A pole 6 m high casts a shadow $2\sqrt{3}$ m long. Angle of elevation of the sun is:

- (A) 30°
- (B) 45°

(C) 60°

(D) 90°

Q14. The angle of elevation of the top of a tower from points P, Q (Q closer to tower) are 30° , 45° . Height of tower if $PQ = 20$ m is:

(A) 20 m

(B) $10(\sqrt{3} + 1)$ m

(C) $10(\sqrt{3} - 1)$ m

(D) $20\sqrt{3}$ m

Q15. Two poles 10 m and 14 m high stand on opposite sides of a road. From the top of the shorter pole, angle of elevation of the top of taller is 45° . Width of road is:

(A) 4 m

(B) 8 m

(C) 6 m

(D) 10 m

Q16. A tree breaks due to storm and the broken part bends so that the top touches ground making 30° with horizontal. Distance of foot of tree from where top touches ground is 8 m. Height of tree is:

(A) 8 m

(B) $8\sqrt{3}$ m

(C) $8(\sqrt{3} + 1)/\sqrt{3}$ m

(D) 16 m (i.e. broken part $16/\sqrt{3} + 8/\sqrt{3} \dots$)

Q17. A balloon at height 30 m makes angle of elevation 60° from a point. Horizontal distance from point is:

(A) $10\sqrt{3}$ m

(B) $30\sqrt{3}$ m

(C) 30 m

(D) 60 m

Q18. From a point on a horizontal plane, angle of elevation of top of a tower is 30° . After walking 60 m towards tower, angle becomes 60° . Height of tower is:

(A) 30 m

(B) $30\sqrt{3}$ m

(C) 60 m

(D) $20\sqrt{3}$ m

Q19. The angle of elevation of the top of a hill from a point P at distance d is 30° . The height of hill is:

(A) $d/\sqrt{3}$

(B) $d\sqrt{3}$

(C) d

(D) $d/2$

Q20. The angle of depression of a car from the top of a tower is 30° . The car is 100 m from the tower base. Height of tower is:

(A) $100/\sqrt{3}$ m

(B) $100\sqrt{3}$ m

(C) 50 m

(D) 100 m

Q21. A ladder rests against a wall making an angle of 60° . Foot is 4 m from wall. Length of ladder is:

(A) 4 m

(B) 8 m

(C) $4\sqrt{3}$ m

(D) $8/\sqrt{3}$ m

Q22. A man observes the top of a tower at angle of elevation θ . If he moves towards the tower the angle of elevation:

(A) Decreases

(B) Increases

(C) Stays same

(D) Becomes zero

Q23. At what angle of elevation will the height of a tower equal its shadow length?

(A) 30°

(B) 45°

(C) 60°

(D) 90°

Q24. From the top of a 75 m high lighthouse, angle of depression of a boat is 30° . Distance of boat from base is:

(A) 25 m

(B) 75 m

(C) $75\sqrt{3}$ m

(D) $50\sqrt{3}$ m

Q25. In a right triangle with legs h and d, height/distance ratio = $\tan \theta$ where θ is:

(A) Angle of depression

(B) Angle of elevation

(C) Both A and B

(D) Neither